DATA MINING AND MACHINE LEARNING 1 PROJECT REPORT

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Abstract: This Data mining and machine learning project report is based on 5 different classification and Regression models which will be applied to 3 datasets from each different sector. The datasets contain the information based on online sales of the home and office supplies of different regions of the U.S, data for 10,000 courses development category from Udemy website and historic data of weather of Australia where the prediction of rain is given. Five models are selected to apply to these datasets. Multiple Regression and Support vector machine for Superstore dataset to predict future sales. Decision tree and Multiple linear regression on Udemy dataset to predict the average ratings if any new book is published and Logistic and Random forest as classification problem to predict forecast of Rain on the next day as Yes/ No. KDD methodology will be first applied to the datasets to clean the datasets and extract the important and useful features. By using KDD and different models, we will calculate the good accuracy to predict the values.

Keywords: Classification, Regression, R2, RMSE, NRMSE, Accuracy, confusion matrix

I. INTRODUCTION

In the current situation, data plays a vital role from small to large organizations to predict future business in terms of growth and money. Using historic data one can predict the business and different factors and features that will affect the business. A most important factor for the organization is the analyze the data and find out the factors responsible which will affect the growth to increase as well as the factors which affect decrease the growth.

By using data mining and machine learning technique, it is possible to analyze the data. Data mining is the process to study the data and extract out the features which are responsible for calculating analysis. This technique is carried out on chunks and millions of data for analysis.

Over the years E-commerce is in trend for buying and selling products. Organizations need to keep track of the sales and the factors affecting the sales over time. For any e-commerce company, the main slogan is to increase sales, so it is important to identify the correct feature to increase sales. We have another dataset that includes the data for udemy courses. Creating content for online courses is very time-consuming and needs lots of work. In today's world when a person checks for any online content one checks for the reviews and ratings. It is very important to predict the ratings from the historic data whether the published content or course will get good ratings or not. After that, we have a weather dataset that includes the weather conditions for the country Australia. Forecasting weather is no easy task. Predicting climatic conditions takes lots of effort. By using historic data, we will try to predict whether it is going to rain tomorrow or not.

II. RELATED WORK

In 2018 Gopal Krishna T, published a research [1] on Superstore data. They were having historic sales data from year 2011- 2013. They tried to predict sales for year 2014. The model applied on the data was Linear Regression model to predict the sales. Precision and accuracy methods are used for evaluating the results. Accuracy rate is 84%. It was found that trust for the general audience is necessary to build for improving profit and sales. Reasearch from Kaggle [2] shows the exploratory analysis of superstore dataset. Data set is analysed and plots are created in order to understand the data in more clear way. Research[3] conducted by Mr.Sai Nikhil Boyapati shows the working of random forest model to predict sales. Only difference for this research project is prediction was done on the food sales and not on Superstore sales. But by analyzing the research different algorithms were studied and implemention of the algorithms were analyzed. In next research[4], sales was forecasted by using python. Different algorithms were used to carry out predictive analysis. In next research [5] prediction is done on ratings. Naïve Bayes models are applied to carry out research with Multi-nomial model. In this research both the models shows same amount of accuracy. Different correlation plots are studied in this. Model selection by using features is the key feature to select research. Following research [6] shows the analysis on ratings of movies database. Classifications models are used to predict ratings and reviews. Continuous variables are transformed in to different levels and result is predicted for the analysis.In this research[7] study of decision trees are analyzed. Classification problems for the forecasting is used to analyze models.It is found that 3 decision tree algorithms are used namely 1D3, CART 4.0 and C4.5. Out of these C4.5 gives best accuracy for the decision trees. Methods are evaluated by using Gini Index.

III. METHODOLOGY

This part of the section is the most important and the crucial section to predict the better results and improve the accuracy for any model. It is very important to prepare and process the data to analyze machine learning and Statistical models. There are variety of data mining methodologies and techniques which are practiced by data scientist. For this project Knowledge Discovery in Database (KDD) data mining method will be used [8]. The first approach is to understand the application domain of the data. In this research three datasets are selected, online sales of the home, office and electronic equipment. Second dataset include data of online content from the Udemy websites and third include the daily forecast of Australian weather. Second step include selecting the target variable for predictions and the independent variables with strong correlation on target variable. Once the target variable is selected we cleaned the data i.e. Data cleaning and pre-processing. Data cleaning and pre-processing is the most important and crucial step in Data mining. Some general steps are used to follow the process. These steps [9] includes.

- Acquiring the dataset from data source
- Importing the important libraries
- Importing the datasets from source
- Identifying missing values and handling them
- Converting categorical variables
- Splitting data
- Feature scaling data

Once the data pre-processing is done, type of problem is identified. This include choosing the correct data mining task i.e. Classification, regression or clustering. Once the problem is identified next step is to identify a model. Model is selected by analyzing data and type of problem, Model is evaluated and applied on training and test data. Prediction is done and model is evaluated. Below Fig.1 shows the steps for KDD approach.

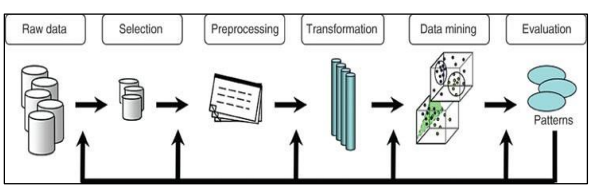

Fig 1 – KDD Process

IV. IMPLEMENTATION

A. *Superstore data*

- Data Selection: Superstore data is extracted from the source: https://www.kaggle.com/bravehart101/sample-supermarket-dataset

Superstore data contain the records of the online sales of the Home, office and Electronic products. Dataset contains 10004 rows and 13 columns. Predictive analysis is done on the data and Sales is identified as the dependent variable. As sales is continuous variable Regression problem is identified and regression models are used.

- Data cleaning and pre-processing: The first step to clean the data is to find the missing values. Missing values are checked by using **sapply()** function. Using **missmap()** function from "Amelia" package, map is created to compare observed values and missing values in Fig.2.

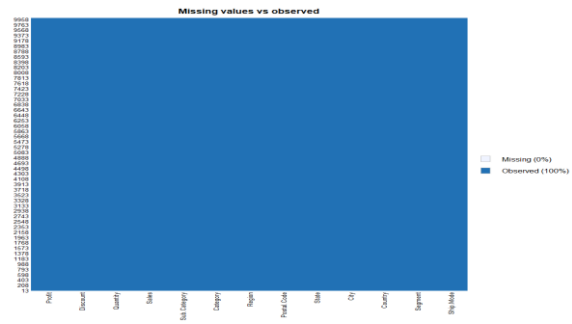

Fig 2 – Missing values (Observed vs Missing)

From the Fig.2, it was found that there were no missing values in the data. Later, data was checked for the outliers. Outliers were checked by using boxplots and Rosner's test. Rosner's test is generalized test to detect the outliers. Rosner's test is executed by using rosnerTest() function from "EnvStats" package. 30 outliers were detected in column 'Quantity'. 421 outliers were detected in column 'Sales'.

```
Test Statistic Parameter:        k = 1000

Alternative Hypothesis:          Up to 1000 observations are not
                                 from the same Distribution.

Type I Error:                    5%

Number of Outliers Detected:     421
```

- Data reduction and transformation: In this process outliers are handled and insignificant columns are dropped by checking correlation between independent and dependent variables. Outliers are handled by using one of the feature scaling technique called normalization. Log transformation is applied to normalize the range for the column 'Sales' and handle outliers. Fig. 3 shows the histogram plot where the 'sales' is normalized.

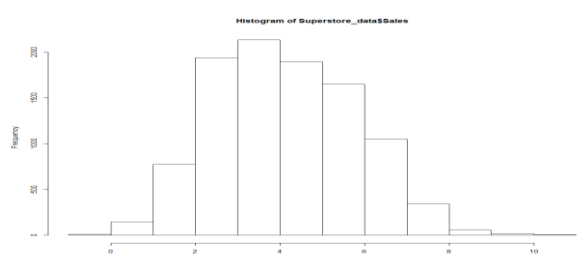

Fig.3 – Histogram for Sales

Once the data is normalized, variables are checked for the correlations. Different correlations are used to compare in order to find meaningful independent variables. On this data 'Pearson' and 'Kendall'

correlations are used. This is done by using rcorr() function from “Hmisc” package. Pearson correlation matrix is shown in the below Fig.4.

	Ship.Mode	Segment	City	State	Postal.Code	Region	Category	Sub.Category	Sales	Quantity	Discount	Profit
Ship.Mode		0.7049	0.7297	0.2573	0.6984	0.0197	0.6899	0.7224	0.8465	0.0226	0.9436	0.5146
Segment	0.7049		0.5826	0.6824	0.3428	0.6814	0.2967	0.2801	0.2891	0.3703	0.0901	0.1843
City	0.7297	0.5826		0.3263	0.0000	0.0000	0.1699	0.5336	0.5746	0.9168	0.0000	0.0150
State	0.2573	0.6824	0.3263		0.0000	0.0000	0.5452	0.8454	0.6765	0.8820	0.0000	0.0115
Postal.Code	0.6984	0.3428	0.0000	0.0000		0.0000	0.8302	0.7279	0.0164	0.1819	0.0000	0.0026
Region	0.0197	0.6814	0.0000	0.0000	0.0000		0.6227	0.3644	0.6848	0.1570	0.0000	0.0266
Category	0.6899	0.2967	0.1699	0.5452	0.8302	0.6227		0.0000	0.0000	0.7102	0.0000	0.0000
Sub.Category	0.7224	0.2801	0.5336	0.8454	0.7279	0.3644	0.0000		0.0000	0.0718	0.0000	0.0030
Sales	0.8465	0.2891	0.5746	0.6765	0.0164	0.6848	0.0000	0.0000		0.0000	0.0049	0.0000
Quantity	0.0226	0.3703	0.9168	0.8820	0.1819	0.1570	0.7102	0.0718	0.0000		0.3927	0.0000
Discount	0.9436	0.0901	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0049	0.3927		0.0000
Profit	0.5146	0.1843	0.0150	0.0115	0.0026	0.0266	0.0000	0.0030	0.0000	0.0000	0.0000	

Fig 4. Correlation matrix

Column ‘Country’ is dropped as it was carrying not much significance with single value. Required features were encoded using as.( ) function to fit the model. Data is split in to 75% as train and 25% as test to predict the values.

- Model selection: In this process model is selected according to the problem. We have identified the problem as regression problem as the dependent variable ‘Sales’ have continuous values. Multiple Linear Regression and Support Vector Regression (SVR) are used to predict the ‘sales’. 5 independent variables are used to predict the sales.

a) Multiple Linear Regression: Multiple Linear Regression is the regression model and linear approach to predict the continuous values of dependent variable by using multiple independent variables. The methodology helps analysts to assess the model variation and the relative contribution of the total variance of each independent variable. MLR is applied on training data and values will be predicted on test data.

b) Support Vector Regression: Support Vector Machine includes unsupervised learning models which are used for both classification and regression problems. SVR is used for the regression problems to predict the quantitative and continuous variable. SVR model is applied on 80% of train data and compared with test data to predict the values and accuracy of the models.

- Evaluation and Interpretation: Both MLR and SVR model are executed and analyzed. Different evaluation methods for each of the models are evaluated and concluded.

a) Multiple Linear Regression: Multiple Linear Regression is the regression model and linear approach to predict the continuous values of dependent variable. To apply Multiple Linear Regression model, we have dependent variable as sales and independent variables as Region, discount, Quantity, Category and Sub-category. Model is applied by using **lm()** function. By applying the model, from model summary we can explain the residuals from below description.

```
Residual standard error: 1.066 on 9981 degrees of freedom
Multiple R-squared:  0.5799,    Adjusted R-squared:  0.579
F-statistic:   656 on 21 and 9981 DF,  p-value: < 2.2e-16
```

From the summary above multiple conclusions can be carried out. F-Statistics shows weather the linear model for the data fits better or not with no independent variables. From the given ANNOVA table we can find that $p < 0.05$, which concludes that regression model fits better.

```
> rmse_ML
[1] 2.060984
> ML_nrmse <- nrmse(pred,test$Sales)
warning message:
In x[m] - y[m] :
  longer object length is not a multiple of shorter object length
> ML_nrmse
[1] 1.646752
> #Evaluation
> actual <- Superstore_data$Sales
> predicted <- pred
> R2 <-1-(sum((predicted-actual)^2)/sum((actual-mean(actual))^2))
> R2
[1] 0.5798802
```

RMSE and NRMSE can from “Metrics” package can be used to evalue the results.

The above figure description shows RMSE (Root mean Squared Error) as 2.06. In theory, it is said that for data between range between 0 to 1000,0.7 value for RMSE is considered to be good value, so as the data set increases there is increase in RMSE value. For which normalizing RMSE i.e. NRMSE can be used to calculate value.

R2 shows the proportion of variance between independent variables for the model. As independent variables increases R2 also increases. R2 is considered good if it falls in range between 0 and 1. Model is considered better if R2 deflects towards 1. As R2 for Multiple regression is 0.57, our model is considered good fit.

b) Support Vector Regression: SVR is model used to predict classification and Regression problem. To apply SVR, we will use svm() function from “e1071” package. Same independent and dependent variables which are used for Multiple Regression will be used for SVM.

After implementing SVM Model following reading from model summary can be concluded.

```
Parameters:
   SVM-Type:  eps-regression
 SVM-Kernel:  radial
       cost:  1
      gamma:  0.04166667
    epsilon:  0.1

Number of Support Vectors:  8604
```

SVM_Type is used for SVM regression problem. Gama value of 0.0416 is calculated. Lower gamma values implies more influence.

```
> R2_SVM
[1] 0.5848688
> rmse_ML <-  rmse(pred1, test$Sales)
warning message:
In actual - predicted :
  longer object length is not a multiple of shorter object length
> rmse_ML
[1] 2.137881
> Super_rmse <- nrmse(pred1, Superstore_data$Sales)
> Super_rmse
[1] 0.7723615
```

In above description shows RMSE value as 2.137 and it is normalized using nrmse() function where the value

is 0.77. R2 value is 0.584 which shows that SVM model is good fit for data to predict sales.

B. Udemy dataset

- Data Selection: Udemy data is selected from the data source : https://www.kaggle.com/jilkothari/finance-accounting-courses-udemy-13k-course

  Udemy dataset contains the data for online content courses of about 10,000 financial courses. The dataset contains 13609 rows and 14 columns. avg_ratings is considered to be a dependent variable. From this dataset we will try to predict how much rating course will get, if the new course is published. avg_rating will be used as dependent variable. As the response variable is continuous and quantitative, regression model will be applied on the dataset.

- Data cleaning and pre-processing: Data is in the form of .csv format. It is first loaded in to R using read_csv() function. Insignificant columns are dropped from the dataset. Once the data set is loaded in to R, it is checked for the missing values using **sapply()** function. Missing values are detected in the data.

```
> sapply(udemy_data_dropped,function(x) sum(is.na(x)))
            is_paid      num_subscribers           avg_rating          num_reviews num_published_lectures
                  0                    0                    0                    0                    0
discount_price__amount              price
               1403                  497
>
```

Columns 'discount_price_amount' and 'price' contains 1403 and 497 missing values. missmap() function is used to plot the percent of missing values from package "Amelia".

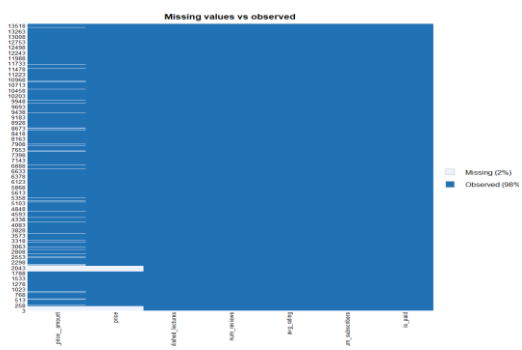

Fig.5 – Missing vs Observed

Fig.5 shows both the columns contains of total 2% of missing values. These values are handled by using the imputation technique. There are different methods which are used to impute mean. On this dataset 'Predictive Mean Matching' imputation technique from package 'mice' is used to find the pattern and handle missing values. At first md.pattern() function is used to find the patterns. Fig.6 below shows the pattern for the missing values.

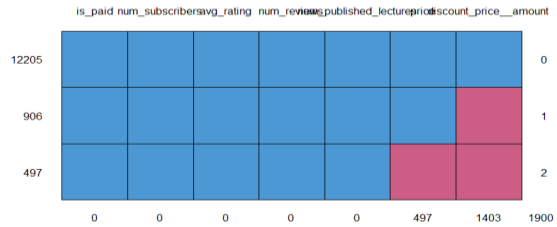

Fig.6 – Pattern for Missing values

Once the pattern is identified, by using aggr() function from "VIM" package is used for each of the column to plot containing missing pattern. Fig.7 below shows the percentage of missing value for each of the columns.

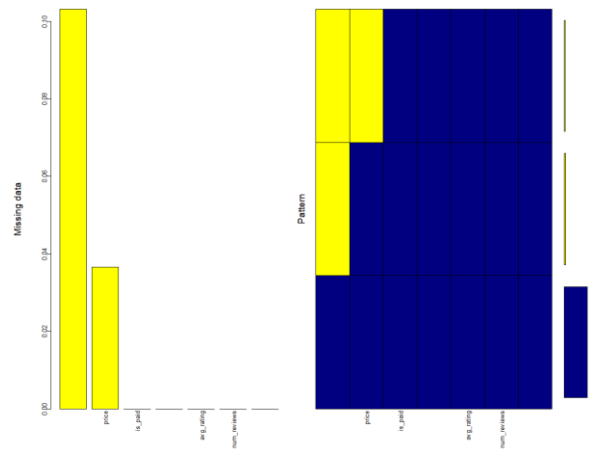

Fig.7 – Missing values pattern for columns

It is important to understand the pattern, pattern decides imputation techniques to be used on the dataset. After computing the pattern, missing values are imputed using "Predictive Mean Matching" technique [10] [11]. PMM is a technique to calculate multiple imputations by giving value to 'm' in the function. 'mice' function from "mice" package is used to impute values. If m=5, then this function will calculate 5 imputed data values for missing values. Imputed values can be checked individually by calling the column. Observed values and imputed values can be observed by using density plot. Fig.8 below shows the density plot for observed against imputed values for m=5. These imputed values are substitute in to missing values by using complete() function.

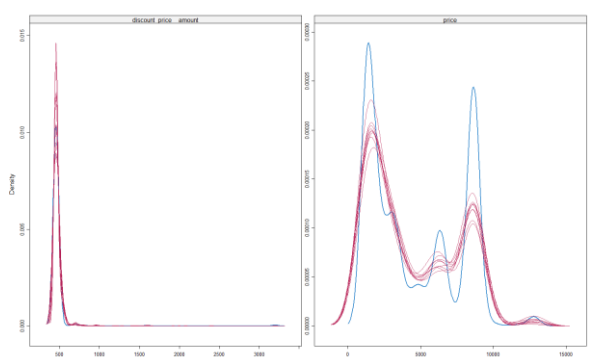

Fig. 8 – Imputed mean (Observed vs Imputed)

Fig.9 below shows the zero missing values checked for the dataset after imputation.

```
> sapply(Udemy_newdata,function(x) sum(is.na(x))) # 0 missing values replaced with mean
              is_paid       num_subscribers            avg_rating           num_reviews num_published_lectures
                    0                     0                     0                     0                    0
discount_price__amount                 price
                    0                     0
```

Fig.9 – Missing values

Now that the missing values are taken care of, outliers are checked for the out of range values. Outliers are checked by using boxplots and Histograms. To check the outliers boxplot() and hist() functions are used. Fig. 10 and Fig.11 below shows boxplot and histograms for entire dataset.

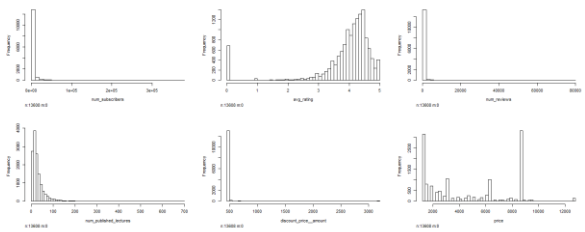

Fig 10 – Histogram for Udemy data

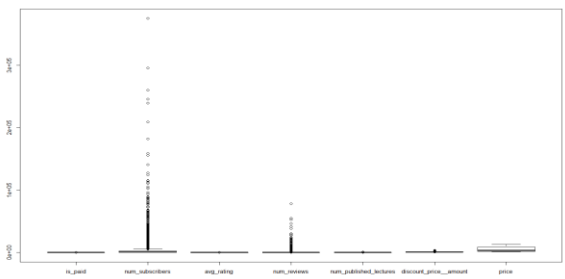

Fig.11 – Boxplots for Udemy data

From the figures, it is analyzed that column 'num_subscribers' and 'num_reviews' contains outliers which needs to be handled.

- Data reduction and transformation: In this process outliers are handled. This is handled by using normalization technique. For the outliers to handle log normalization technique has been used. This is done by applying log on columns with outliers. After applying the log the data is normally distributed and outliers are handled. Fig. 12 shows the boxplot after handling outliers for Udemy dataset.

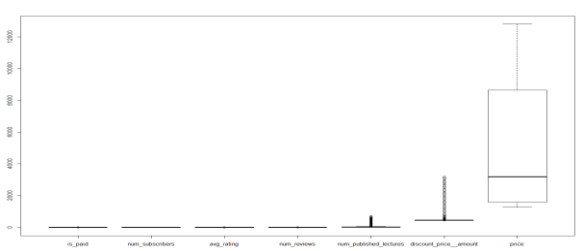

Fig. 12 – Boxplot for Udemy data after handling outliers.

Outliers for both columns are handled using Normalization technique. Next process is to find the correlation matrix for variables but, it was analyzed as column 'price' have the values which are not matching with the range of other variables. This can affect the model when applied on such data. Thus column 'price' is normalized using same log function in order to find correlation matrix falling within the same range. This is done applying log() function on the column. This normalize the range from 8000 - 10000 to 7 – 10, moving forward to print correlation matrix to find relation between dependent and independent variables. To print the correlation "kendall" correlation method is used. Fig. 13 and Fig.14 shows the correlation matrix and heatmap for the dataset. This is done by using cor() function and heatmaply() function from "heatmaply" package.

```
> Heatmap_corr <- cor(Udemy_Correlation, use="complete.obs", method ="kendall") #Storing correlation matrix in dataframe
> Heatmap_corr
                      is_paid num_subscribers avg_rating num_reviews num_published_lectures discount_price__amount
is_paid            1.00000000    -0.111595993 0.01671857 -0.05415734            0.09749216          -0.002398730
num_subscribers   -0.11159599     1.000000000 0.08182195 0.58887308            0.19585108           0.005993469
avg_rating         0.01671857     0.081821950 1.00000000 0.16871500            0.13680055           0.059413859
num_reviews       -0.05415734     0.588873076 0.16871500 1.00000000            0.23167328           0.064968223
num_published_lectures 0.09749216 0.195851079 0.13680055 0.23167328            1.00000000           0.044380980
discount_price__amount -0.00239873 0.005993469 0.05941386 0.06496822            0.04438098           1.000000000
price              0.01464653     0.147676993 0.10536872 0.18192928            0.23196710           0.139302505
                        price
is_paid            0.01464653
num_subscribers    0.14767699
avg_rating         0.10536872
num_reviews        0.18192928
num_published_lectures 0.23196710
discount_price__amount 0.13930251
price              1.00000000
```

Fig.13 – Correlation matrix (Udemy data)

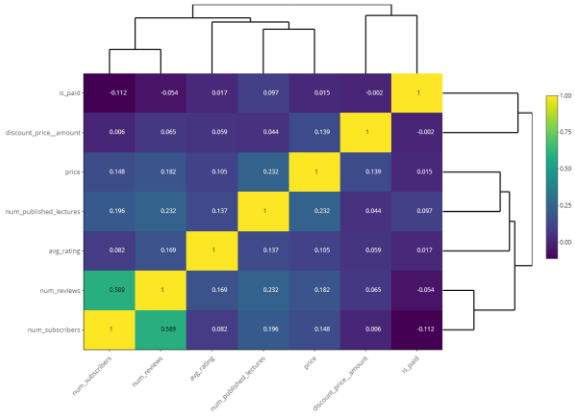

Fig. 14 – Correlation heatmap (Udemy data)

By checking correlation matrix insignificant columns are dropped. Data conversion as factors are done by using as.factor() function. Once data is ready for the model, data is split in to train and test. Training data contains 75% and test data contain 25% data.

- Model Selection: In this process, model is selected by analyzing data. As analyzed the problem is detected as Regression problem so we will select two regression models to predict the dependent value. We have dependent value as "avg_rating" where average rating for upcoming published content will be predicted by using 4 independent variables. The model selected to predict is decision tree for regression.

  a) Decision tree: Decision tree [12] is a predictive model used for classification and regression problems. Decision tree takes the target variables as regression trees. For classification method= class is used to predict and to predict regression

values method=annova [5] is used to predict the values.

- Evaluation and Interpretation: Decision tree regression is applied on training sets by using rpart() function from "rpart" package and values are predicted on test set. Different Evaluation methods will be used to analyze the model accuracy and evaluate errors. Decision tree works on tree and node approach.

```
> R2_dt <- 1 - (sum((predicted_u-actual_u)^2)/sum((actual_u-mean(actual_u))^2))
> R2_dt
[1] 0.4342675
> rmse_ML <-  rmse(ba,test$avg_rating)
> rmse_ML
[1] 1.181633
> Normal_rmse <- nrmse(ba,test$avg_rating)
> Normal_rmse
[1] 2.021662
```

To evaluate the model R2 value is checked. R2 value is 0.43 i.e. model explain only 43% of variance. RMSE and NRMSE is also calculate.

C. Weather dataset

- Data selection: Australian weather dataset is acquired from the source: https://www.kaggle.com/trisha2094/weatheraus

  Australian weather data contains the data for daily climatic conditions for Australia weather and forecast weather next day it is going to rain or not. Data contains 142194 rows and 19 columns. In this data, predictor will be "RainTomorrow" column, which will be considered as dependent variable. The dependent variable has two levels Yes/No, so this is considered as classification problem and classification models will be applied to predict the values.

- Data cleaning and pre-processing: In this data, first step is to delete unwanted columns. There lies some columns which for modelling are actually not required for time being. Hence those columns will be dropped. Next step is to find the missing values in the dataset. This is done by using sapply() function. Fig.15 shows the missing values for each of the column.

```
> sapply(weatherdropped,function(x) sum(is.na(x)))
    MinTemp     MaxTemp    Rainfall Evaporation    Sunshine WindSpeed9am WindSpeed3pm  Humidity9am  Humidity3pm  Pressure9am
        637         322        1406       60843       67816        1348        2630         1774         3610        14014
Pressure3pm    Cloud9am    Cloud3pm     Temp9am     Temp3pm RainTomorrow
      13981       53657       57094         904        2726            0
>
```

Fig.15 – Columns containing missing values

By using missmap() function from package "Amelia" map is plotted which shows total percentage of missing values against observed values. Fig.16 shows 12% missing values in the data.

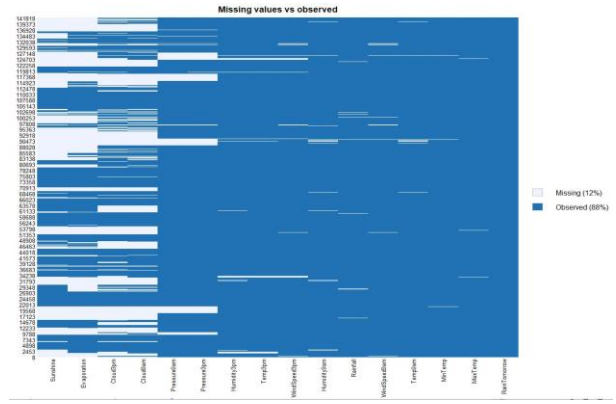

Fig.16 – Missing vs Observed values

Handling missing values is the most important step in any data mining technique. For weather data missing values are handled by using 'pmm' method [10] [11] i.e. predictive mean matching. In predictive mean matching values are imputed by calculating mean. mice() function is used from 'mice' package to impute mean. In function mice value of m is 2, where two possible imputations will extracted. Fig.17 shows the density plot to for observed data and two data values of imputed mean.

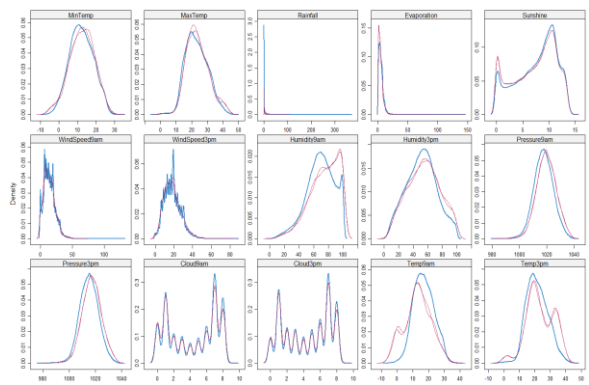

Fig.17 – Density plot for Imputed means, m=2

To continue with the further tests, imputed mean is taken in to missing values. Since we imputed missing values, missing values are again checked and from Fig.18 we conclude that there are no missing values in the dataset.

```
> sapply(weather_new_data,function(x) sum(is.na(x))) # 0 missing values replaced with mean
    MinTemp     MaxTemp    Rainfall Evaporation    Sunshine WindSpeed9am WindSpeed3pm  Humidity9am  Humidity3pm  Pressure9am
          0           0           0           0           0           0           0            0            0            0
Pressure3pm    Cloud9am    Cloud3pm     Temp9am     Temp3pm RainTomorrow
          0           0           0           0           0           0
>
```

Fig.18 – Zero missing values for dataset

Next step includes checking outliers in data. Handling outliers will predict better accuracy models. Outliers are checked using boxplots and histograms. Fig.19 shows boxplot for the data. It is seen that there are not much outliers which lies in the data.

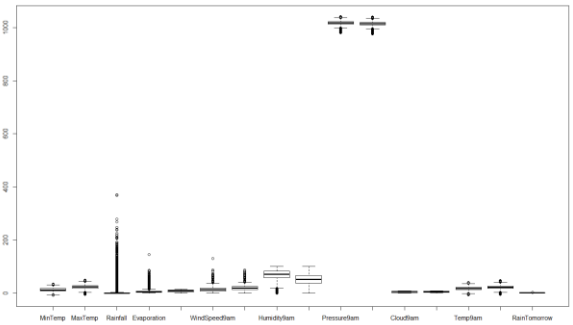

Fig.19 – Boxplot for Weather data

Only thing we conclude that the range for the 'Pressure' is not matching with other variables. Range for the pressure is handled by using transformation technique and data is normalized to check for correlations.

- Data reduction and transformation: In this section data is normalized within a particular range. Normalization is a technique used for feature selection. In this dataset log normalization is done on Pressure9am and Pressure Pressure3pm using log() function. Compared to Fig. the values range for Pressure is more than 800. Fig.20 shows boxplot for data which lies within a range i.e. between 0 to 300 and can be used to extract independent variables for model prediction.

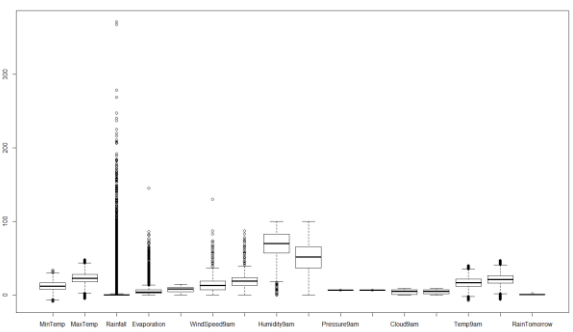

Fig.20 – Boxplot after Normalization

Once the data is ready correlations are checked among variables to extract independent variables. Correlation between the variables are checked by using corrplot() function from "corrplot" package. Fig.21 and Fig.22 shows the correlation matrix for dataset and plot for correlation matric

```
> Correlation_matrix <- corrplot(cor(weather_correlation), method = "number")
> correlation_matrix
              MinTemp     MaxTemp    Rainfall Evaporation   Sunshine WindSpeed9am WindSpeed3pm Humidity9am Humidity3pm
MinTemp     1.00000000  0.73647340  0.10390417  0.47222558  0.08596441  0.17544193  0.17336974  -0.2373032  0.00314137
MaxTemp     0.73647340  1.00000000 -0.07442862  0.59397241  0.48140752  0.01359377  0.04856615  -0.5074499 -0.50948015
Rainfall    0.10390417 -0.07442862  1.00000000 -0.05439112  -0.23647278  0.08769702  0.05773073   0.2239141  0.25551407
Evaporation 0.47222558  0.59397241 -0.05439112  1.00000000  0.34228806  0.19897588  0.14003527  -0.5200128 -0.40640560
Sunshine    0.08596441  0.48140752 -0.23647278  0.34228806  1.00000000 -0.01033553  0.05453695  -0.4840562 -0.64449569
WindSpeed9am 0.17544419 0.01359377  0.08769703  0.19897587 -0.01033553  1.00000000  0.52115754  -0.2694625 -0.02981465
WindSpeed3pm 0.17336974 0.04856615  0.05773073  0.14003527  0.05453695  0.52115753 1.00000000  -0.1458876  0.01689729
Humidity9am -0.23730319 -0.50744991  0.22391412 -0.52001280 -0.48405620 -0.26946264 -0.14588755  1.0000000  0.66875283
Humidity3pm  0.00314137 -0.50948015  0.25551407 -0.40640560 -0.64449569 -0.02981465  0.01689729  0.6687528  1.00000000
Pressure9am -0.45894984 -0.34198941 -0.16347649 -0.27103638  0.02584666 -0.22233119 6 -0.30205631  0.1440659 -0.01205750
Pressure3pm -0.47261918 -0.43898248 -0.12195844 -0.29263143 -0.03341545 -0.16871549 5 -0.26129576  0.1912332  0.06955993
Cloud9am    0.08496511 -0.29440333  0.19815061 -0.19256253 -0.68213815  0.03626564  0.05220153  0.4309129  0.51697158
Cloud3pm    0.03032560 -0.28423161  0.17370580 -0.19078223 -0.70968039  0.06715907 6 0.03153840  0.3365582  0.52045100
Temp9am     0.90225394  0.88791376  0.01004709  0.55400188  0.30339532  0.12631892 3  0.16170288 -0.4731730 -0.22624346
Temp3pm     0.71170423  0.98470789 -0.07836904  0.57534958  0.50230115 -0.00170874 9  0.02282845 -0.4980945 -0.55810877
RainTomorrow 0.08416503 -0.15935228  0.23907514 -0.11928311 -0.46037272  0.09145764 5  0.08782485  0.2578104  0.44752481
             Pressure9am Pressure3pm    Cloud9am    Cloud3pm     Temp9am     Temp3pm RainTomorrow
MinTemp     -0.45894984 -0.47261918  0.08496511  0.03032560  0.90225394  0.71170423  0.08416503
MaxTemp     -0.34198941 -0.43898248 -0.29440333 -0.28423161  0.88791376  0.98470789 -0.15935228
Rainfall    -0.16347649 -0.12195844  0.19815061  0.17370580  0.01004709 -0.07836904  0.23907514
Evaporation -0.27103638 -0.29263143 -0.19256253 -0.19078223  0.55400188  0.57534959 -0.11928311
Sunshine     0.02584666 -0.03341545 -0.68213815 -0.70968039  0.30339532  0.50230115 -0.46037272
WindSpeed9am -0.22233120 -0.16871550  0.03626564  0.06715908  0.12651892 -0.00170874 9  0.09145764
WindSpeed3pm -0.30205631 -0.26129576  0.05220153  0.03153840  0.16170288  0.02282845  0.08782485
Humidity9am  0.14406588  0.19123319  0.43091286  0.33655818 -0.47517296 -0.49809446 4  0.25781037
Humidity3pm -0.01205750  0.06955993  0.51697158  0.52045100 -0.22624346 -0.55810877  0.44752481
Pressure9am  1.00000000  0.96164609 -0.12148773 -0.14264336 -0.43066681 -0.30026872 4 -0.24373280
Pressure3pm  0.96164609  1.00000000 -0.05380500 -0.08013215 -0.48139743 -0.40622116 8 -0.22330425
Cloud9am    -0.12148773 -0.05380500  1.00000000  0.61302131 -0.13830464 -0.31153470 8  0.31339885
Cloud3pm    -0.14264336 -0.08013215  0.61302131  1.00000000 -0.12465849 -0.32317561 1  0.36978734
Temp9am     -0.43066681 -0.48139743 -0.13830464 -0.12465849  1.00000000  0.86287601 7 -0.02734952
Temp3pm     -0.30026872 -0.40622117 -0.31153471 -0.32317561  0.86287602  1.00000000 -0.19327083
RainTomorrow -0.24373280 -0.22330425  0.31339885  0.36978734 -0.02734952 -0.19327083  1.00000000
```

Fig.21 – Correlation matrix

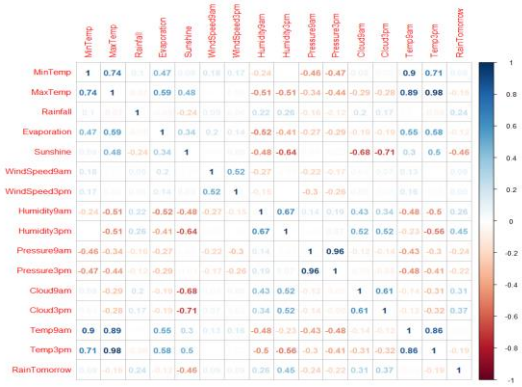

Fig.22 – Correlation plot

Independent variables are extracted from the data with strong relation with dependent variable. In total 5 independent variables will be used to predict weather it's going to rain tomorrow or not. Further, data is split in to training and testing data. 75% data is kept for training model and model will be test on testing data to predict forecast of Rain.

- Model selection: From the above pre-processing steps, problem is identified as classification problem. Two classification models will be applied on the data. Logistic Regression and Random Forest. Total 5 independent variables will be used to predict "RainTomorrow" variable. Description of models are as follows:

a) Logistic Regression: Logic Regression is the most traditional and community model used mainly for classification problems. It is used to predict the class probability. Logistic Regression can also be used for regression problems where dependent variable is divided in to multiple class converting it in to Multi-nominal Logistic Regression. In this research Binary Logistic Regression is applied to model as it contains Binary predictor values.

b) Random Forest : In random forest trees are created by using multitude at training time and output are predicted by using mean or average of individual trees. Random forest can be used for Classification as well as Regression problems.

- Evaluation and Interpretation: Logistic Regression model and Random forest is used to model the data.

a) Logistic Regression: Logistic Regression is used to predict this classification problem. Logistic Regression is applied by using glm() function from "CRAN" package. family=binomial and link = logit is used for Binary classification and Binary logistic regression. Dependent variable is taken as RainTomorrow and Independent variables are taken as Rainfall, Sunshine, Humidity3pm, Cloud9am and Cloud3pm. For logistic Regression Accuracy is calculated by using prediction() and performance() function from "ROCR" package. Following output is calculated from these model.

```
> auc
[1] 0.8357981
> ################ Confusion matrix #################
> # Printing confusion matrix
> Con_mat_log <- table((weather_model$RainTomorrow), pred > 0.5)
> Con_mat_log

       FALSE   TRUE
  No  107749   2567
  Yes  21852  10025
>
```

auc shows the accuracy of model. We can see that 83% of the Accuracy for model. Below that is printed Confusion matrix. From the confusion matrix we can conclude that 107749 True negative values were predicted 2567 False positive, 21852 False negative and 10025 True positive are predicted, showing Type I error.

b) Random Forest : Random Forest model is used for Classification and Regression problems. Random forest is executed by using randomForest() function from "randomForest" package. Number of independent variables are same which are used for Logistic Regression which is executed above. Following conclusions can be interpreted by using this model.

```
Call:
 randomForest(formula = weather_model$RainTomorrow ~ weather_model$Rainfall +      weather_model$Sunshine + weather_model$H
 + weather_model$Cloud9am +      weather_model$Cloud3pm, data = weather_model)
               Type of random forest: classification
                     Number of trees: 500
No. of variables tried at each split: 2

        OOB estimate of  error rate: 16.9%
Confusion matrix:
        No   Yes class.error
No  103825  6491  0.05884006
Yes  17545 14332  0.55039684
>
```

Above description shows the confusion matrix. Values shows 17545, false positive and 6491 as false negative values, showing 55% Type I error and 5% Type II error. Only 40% of the values are predicted correctly.

V. CONCLUSION

From the analysis we can conclude that, on Superstore database Multiple Regression and Support vector machine works best on Superstore data to analyze sales and predict values for sales. Selecting features is important in superstore data. In a same way for Udemy dataset decision tree is not much friendly approach, different other approaches can be applied to predict more accurate outputs. From Weather data we conclude that, to forecast weather applying classification models can be tricky. More missing values can affect model badly. Using Random forest and Logistic regression is traditional approach to forecast the response.

VI. FUTURE WORK

Future wok includes working on Weather dataset. Multiple columns were used to predict the response. Thus idea of dimension reduction came in to picture. Different climatic conditions of same day according to the time can converted in to single component. Thus reducing dimensions. Weather data is based on daily basis thus there are chances of Time series analysis will work better for better accuracy.

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